

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I B. Greenitha - 192312203 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B. Greenitha
Signature of the student:

18
20

③

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

① Smart Home automation system

Ans

Smart Home automation system manage lighting, temperature, and security based on the user behaviour and environmental conditions.

* Simple Reflex Agent

- works only on the current input using condition action rules.
- Ex:- If model is detected → Turn on lights
If temperature above 30°C → Turn on AC

Advantages :- Fast and simple

Disadvantage :- Does not remember past events or learn user habits.

* Model-based Reflex

- Maintain a internal model of the environment
- Use current sensor data and previous information.
- Ex:- known a room was occupied recently so it keeps lights on even if motion stops briefly.

Advantages :- Better decision in partially observable environment.

Disadvantages :- Most complex than a simple reflex agent.

*Goal based Agent

→ choose action that achieve a specific goal.

Ex:-

Goal :- maintain room-temperature at 24°C while saving energy.

Advantage:- Flexible

Disadvantage:- Require more computation utility-Based Agent.

*Utility - Based Agent

→ selects the action with highest overall benefit

→ Balances comfort, energy efficiency

Ex:-

Adjust AC and lighting to maximize comfort while minimizing electricity usage.

Advantages:- make a optimal decision with multiple objectives.

Disadvantage:- ~~utility~~ calculation can be complex.

Best Approach:-

These are the most effective for a Smart Home Automation system.

Reason:- → Learn user habits

→ Achieve the goals such as comfort security.

Robot in an unknown maze using CBS & IDS

4 Problem Statement

Robot is placed in an unknown maze and must reach the exit without knowing the layout.

The robot should explore the maze and find a path.

Uniform Cost Search (UCS)

Uniform cost search expands the node with the lowest cumulative path cost.

Working

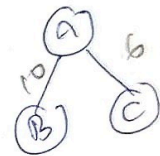
Ex

Paths:

A $\rightarrow$ B $\rightarrow$ Goal = cost 10

A $\rightarrow$ C $\rightarrow$ Goal = cost 6

UCS selects the second path because its total cost is lower.



Advantages

- $\rightarrow$ complete
- $\rightarrow$ Finds the least-cost path.
- $\rightarrow$ Suitable when movement costs vary.

Disadvantages

- $\rightarrow$ High memory usage
- $\rightarrow$ Slow for large mazes

Complexity

Time: $O(b^d (1 + c/\epsilon))^*$

Space: $O(b^d (1 + c/\epsilon))^*$

where

b = branching factor

C^* = optimal path cost

ϵ = minimum step cost

Iterative Deepening search (IDS)

IDS repeatedly performs Depth - Limited search with increasing depth limits.

ex:-

Depth 0

Depth 1

Depth 2

continue until goal is found.

Advantages

- low memory usage
- complete.
- suitable when solution depth is unknown.

Disadvantages

- Repeat node expansion
- Not optimal if step cost differs.

complexity

Time: $O(b^d)$

Space: $O(bd)$

Where

b = branching factor

d = depth of solution

Effective

When compare to the both uniform cost search and Iterative Deepening search. have complete and suitable when solution depth is cost.